

**The FLCL Framework:
Reconstructing the Labour Cost
Transmission
Function in France**

*Average Transmission, Marginal Transmission,
and the Elasticity of Net Income to Employer Cost*

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Abstract. This paper proposes the FLCL (*French Labour Cost to Labour Income*) framework, a functional approach to measuring labour cost transmission in France. Rather than reporting an aggregate tax wedge at a single representative wage point, we reconstruct the full statutory mapping $N = F(C)$ linking employer labour cost to employee net income, and characterise its properties through three analytically distinct objects: the average transmission ratio $A = N/C$, the marginal transmission function $T = dN/dC$, and the net-income-to-labour-cost elasticity $\varepsilon_{NC} = T/A$. A fourth object, the relief transmission premium $R = A - A^0$, isolates the contribution of contribution relief schemes through counterfactual comparison. The framework is implemented in the French Labour Cost Lab dashboard using 50,016 payroll simulations generated from the official Mon-entreprise / URSSAF calculation engine, with parameters as of June 2026. The public dataset covers gross wages from 0.8 to 6.0 times the statutory minimum wage (SMIC = 1,867.02 €/month gross), with a 0.01 SMIC step, across 96 employment profiles combining employee status, territorial regime, firm size, AT/MP scenario, and working-time rate. The analytical results reported in this paper focus on the full-time baseline profile in order to isolate the shape of the statutory transmission function. A central finding is that the French relief system – a combination of the *Réduction Générale des Cotisations Patronales* (RGCP) and the degressive health contribution reduction – generates a zone of depressed transmission extending well beyond the statutory RGCP ceiling: the RGDU (*Réduction Générale Dégressive Unifiée*) analytical zone considered in this paper spans $[1.0, 2.92] \times \text{SMIC}$, not the $[1.0, 1.6] \times \text{SMIC}$ range suggested by the RGCP formula alone. Within this analytical zone, average transmission falls by 17.4 percentage points (from 0.733 to 0.558), marginal transmission drops sharply to 0.419 at the zone entry before recovering gradually, and the elasticity ε_{NC} reaches a minimum of 0.548. These findings reveal an implicit marginal taxation of wage growth spanning nearly three times the minimum wage that is invisible to conventional average tax wedge measures.

Keywords: labour cost transmission, tax wedge, marginal taxation, net-income elasticity, social contributions, contribution relief, minimum wage, French labour market

JEL codes: H22, H55, J32, J38

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The views expressed in this paper are those of the author only.

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1. Introduction

1.1. The Question

Every employment relationship produces two monetary quantities. The firm records a total labour cost C ; the employee receives a net wage N . The ratio N/C – the share of employer expenditure that reaches the worker – is always strictly less than one. The difference, absorbed by mandatory social contributions on both sides of the payroll, has been measured by economists for decades under the heading of the *tax wedge*. Yet the standard implementation, as practised by the OECD in its annual *Taxing Wages* publication (OECD, 2024), compresses an entire function into a scalar: it reports a single wedge value for a worker earning a fixed multiple of the average wage. This approach is communicable and internationally comparable. But it cannot reveal how the wedge changes across the distribution – and in a system as structured as the French contribution schedule, the shape of that variation matters at least as much as its level.

This paper asks instead:

What is the function $N = F(C)$, and what are the properties of that function across the full support of the French wage distribution?

1.2. Motivation: A Relief Zone Much Wider Than Advertised

A striking institutional finding motivates the analysis. The dominant relief scheme in French payroll taxation – the *Réduction Générale des Cotisations Patronales* (RGCP, formerly *allègement Fillon*) – is typically described as degressive between 1.0 and $1.6 \times \text{SMIC}$. Policy discussions, evaluation studies, and most international comparisons treat $1.6 \times \text{SMIC}$ as the upper bound of the relief zone.

The simulation data reveal that this description is incomplete. When the full contribution relief schedule is computed – including both the RGCP proper and the degressive health contribution reduction that was progressively merged into the unified RGDU (*Réduction Générale Dégressive Unifiée*) system – the RGDU analytical zone extends to $2.92 \times \text{SMIC}$, not 1.6 . At $1.6 \times \text{SMIC}$, the monthly relief still amounts to **303 €**; it declines only gradually to zero between 1.6 and 2.92 SMIC . The relief zone therefore spans nearly three times the minimum wage – a reach that no existing indicator documents.

This finding has direct consequences for the empirical profile of transmission. The analytical zone of depressed average transmission considered in this paper is not confined to the $[1.0, 1.6] \times \text{SMIC}$ range of the RGCP formula alone, but extends to $2.92 \times \text{SMIC}$, covering almost the entire lower half of the wage distribution.

1.3. The FLCL Framework

The FLCL (*French Labour Cost to Labour Income*) framework characterises the transmission function F through four objects:

1. $A(C) = N/C$, the **average transmission ratio**: share of employer cost reaching the employee;
2. $T(C) = dN/dC$, the **marginal transmission function**: employee gain per additional euro of employer cost; the first derivative of F ;
3. $\varepsilon_{NC}(C) = T/A$, the **net-income-to-labour-cost elasticity**: percentage change in net income per one-percent increase in employer cost; tests T vs A divergence in a scale-free metric;
4. $R(C) = A - A^0$, the **relief transmission premium**: counterfactual gain in average transmission from the RGDU relief, where A^0 is computed at employer cost without relief.

The central object is $T(C)$. The contribution of this paper is not a new formula for the level of the wedge – A and $1 - \text{OECD wedge}$ are arithmetically equivalent – but the functional reconstruction of F across a dense wage grid enabling the computation of T and ε_{NC} , which are genuinely new empirical objects unavailable from existing publications. In the current public dashboard implementation, this grid contains 521 wage points per profile.

1.4. Relation to the Literature

The paper is adjacent to three bodies of work. The *tax wedge* literature (OECD, 2024; Immervoll, 2007) provides average-level benchmarks; the FLCL framework provides the functional and marginal counterpart. The *tax incidence* literature (Gruber, 1997; Saez et al., 2019; Bozio et al., 2019) asks who bears the burden after equilibrium adjustment; the FLCL framework provides the statutory schedule that serves as input to incidence models. The *French contribution relief* literature (Crépon and Desplatz, 2002; L'Horty, 2000; Carbonnier, 2011) has studied employment effects of the *allègement Fillon*; the FLCL framework reveals the full reach of its successor scheme and quantifies the associated transmission distortion.

1.5. Outline

Section 2 describes the French contribution system and the two-component RGDU. Section 3 presents the dataset. Section 4 develops the theoretical framework. Section 5 presents the empirical results. Section 6 discusses implications and extensions. Section 7 concludes.

2. Institutional Background

2.1. The Architecture of French Payroll Taxation

French labour taxation operates through two channels. On the *employer side*, *cotisations patronales* are levied on gross wages and added to the wage bill, converting gross wages into total employer cost. These cover health insurance, family benefits, two tiers of old-age pension, unemployment insurance, workplace accident insurance (*AT/MP*), and minor levies (FNAL, *versement mobilité*). On the *employee side*, *cotisations salariales* and the CSG/CRDS surcharges are deducted from gross wages, converting them into net wages.

Formally, let $\tau^E(W)$ and $\tau^S(W)$ denote total employer and employee contributions as functions of the gross wage W :

$$C(W) = W + \tau^E(W), \quad (1)$$

$$N(W) = W - \tau^S(W). \quad (2)$$

The elimination of W from (1)-(2) yields the transmission function $N = F(C)$. This paper abstracts from personal income tax, which depends on household-level variables outside the payroll relationship (see Section 6).

2.2. The RGDU Relief System and Its Two-Component Structure

The dominant non-linearity in F is the ensemble of employer contribution relief schemes, operationally captured in the simulation engine as the RGDU (*Réduction Générale Dégressive Unifiée*). The RGDU consolidates two historically distinct instruments:

Component 1 — The RGCP (1.0-1.6 SMIC). The *Réduction Générale des Cotisations Patronales* provides a degressive reduction in employer social contribution rates for wages between 1.0 and $1.6 \times \text{SMIC}$, with the maximum relief at $1.0 \times \text{SMIC}$. In annualised terms:

$$\text{RGCP}(W_a) = \begin{cases} (1 + \bar{\tau}^E) W_a \cdot \frac{T_{\max}}{0.6} \left(\frac{1.6 \text{SMIC}_a}{W_a} - 1 \right) & \text{if } W_a \leq 1.6 \text{SMIC}_a \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

with maximum reduction rate $T_{\max} \approx 0.32$.

Component 2 — The degressive health contribution reduction (1.6-2.5 SMIC).

Since 2019, the health insurance contribution rate for employers has been made degressive above $1.6 \times \text{SMIC}$: firms pay a reduced health rate of 7.0% below $2.5 \times \text{SMIC}$ relative to

the full rate, tapering to zero above $2.5 \times \text{SMIC}$. In the simulation engine, this component is folded into the unified RGDU expression (*salarié.cotisations.exonérations.RGDU*) along with the RGCP.

Empirical reach of the RGDU. The simulation data reveal that the combined RGDU relief is non-zero from $0.8 \times \text{SMIC}$ (the lower bound of our grid) to $2.92 \times \text{SMIC}$, where it abruptly drops to zero. The analytical RGDU zone considered throughout this paper is therefore defined as $[1.0, 2.92] \times \text{SMIC}$, anchored at the statutory minimum wage entry point. Table 1 illustrates the full relief profile, including the sub-SMIC segment reported for completeness.

Table 1: RGDU Relief Profile at Selected Wage Levels

s (SMIC multiples)	Gross (€/m)	Employer cost (€/m)	RGDU (€/m)	A	T
0.80	1,494	1,563	595	0.744	0.768
1.00	1,867	1,988	700	0.733	0.419
1.20	2,240	2,699	522	0.649	0.433
1.40	2,614	3,360	395	0.609	0.461
1.60	2,987	3,986	303	0.588	0.482
1.80	3,361	4,588	234	0.575	0.500
2.00	3,734	5,171	184	0.567	0.513
2.50	4,668	6,582	117	0.558	0.547
2.92	5,452	7,711	109	0.558	0.334
2.93	5,470	7,847	0	0.550	0.335
3.50	6,535	9,369	0	0.552	0.561

Notes: Baseline profile (non-cadre, continental France, standard AT/MP). $\text{SMIC} = 1,867.02$ €/month gross (June 2026). The $s = 0.80$ row is reported for completeness but lies outside the analytical RGDU zone $[1.0, 2.92] \times \text{SMIC}$ used throughout the paper. The abrupt cessation of RGDU at $s = 2.92$ is a coincidence of the PSS ceiling kink with the residual health reduction taper exit.

Three features of Table 1 are immediately notable. First, the RGDU relief at $1.6 \times \text{SMIC}$ (303 €/month) is nearly half its value at $1.0 \times \text{SMIC}$ (700 €/month): the relief does not disappear at the RGCP ceiling but continues, at a reduced level, for another 1.32 SMIC units. Second, the relief profile is non-monotone in absolute terms: it reaches its maximum at $1.0 \times \text{SMIC}$ and declines only slowly in the 1.6-2.9 range (from 303 to 109 €/month over 1.3 SMIC units). Third, the marginal transmission T at the RGDU exit ($s = 2.92$) is 0.334 – the global minimum in the dataset – because the abrupt cessation of 109 €/month of relief coincides with a PSS ceiling kink, creating the most severe local discontinuity in the entire contribution schedule.

Proposition 2.1 (RGCP component taper). Under equation (3) with constant $\bar{\tau}^E$, the RGCP component is linear in W_a over $[1.0, 1.6] \times \text{SMIC}$:

$$\text{RGCP}(W_a) = (1 + \bar{\tau}^E) \frac{T_{\max}}{0.6} (1.6 \text{SMIC}_a - W_a), \quad (4)$$

with slope $-(1 + \bar{\tau}^E) T_{\max}/0.6 \approx -0.71$. Each additional euro of gross wage in this range reduces the RGCP relief by 0.71 €, raising the effective employer marginal cost to 1.71 € per euro of gross wage (versus 1.33 € outside the taper zone), directly depressing marginal transmission T .

Proof. Expand equation (3): $(1 + \bar{\tau}^E) W_a \cdot (T_{\max}/0.6) (1.6 \text{SMIC}_a/W_a - 1) = (1 + \bar{\tau}^E) (T_{\max}/0.6) (1.6 \text{SMIC}_a - W_a)$, which is linear with the stated derivative. $\square$ $\square$

2.3. Special Contribution Regimes

The **Alsace-Moselle** regime adds approximately 1.5 percentage points to the employee contribution rate; the **cadre** (executive) profile entails a higher AGIRC-ARRCO pension tier raising employer contributions by 2-3 percentage points; the **AT/MP** workplace accident rate varies by risk scenario; firm-size thresholds affect selected employer-side contributions; and working-time rates affect the relationship between actual monthly pay and full-time-equivalent SMIC multiples. The current public dataset covers 96 profile combinations.

3. Data

3.1. Simulation Engine and Dataset

All data are generated from *Mon-entreprise.urssaf.fr*, the open-source payroll simulation platform maintained by the URSSAF network, using regulatory parameters as of **June 2026**. The current public dataset contains **50,016 observations**: gross wages are expressed as full-time-equivalent SMIC multiples from 0.8 to $6.0 \times \text{SMIC}$ with step $h = 0.01 \times \text{SMIC}$, yielding 521 wage points per profile, crossed with 96 employment profiles.

The profile dimensions are: employee status (non-cadre/cadre), territorial regime (standard/Alsace-Moselle), firm size (< 50 employees/ ≥ 50 employees), AT/MP risk scenario, and working-time rate (full-time, 80%, 50%). The baseline analytical profile used throughout the results section remains the full-time non-cadre employee under the standard territorial regime, in a firm with fewer than 50 employees and standard AT/MP assumptions.

3.2. Constructed Variables

For each observation (s, φ) , with $s = W/\text{SMIC}$, we construct:

$$A(s; \varphi) = N/C, \quad (5)$$

$$T(s; \varphi) \approx \frac{N(s+h) - N(s-h)}{C(s+h) - C(s-h)}, \quad (6)$$

$$\varepsilon_{NC}(s; \varphi) \approx \frac{\ln N(s+h) - \ln N(s-h)}{\ln C(s+h) - \ln C(s-h)}, \quad (7)$$

$$C^0(s; \varphi) = C(s; \varphi) + \text{RGDU}(s; \varphi), \quad (8)$$

$$R(s; \varphi) = A(s; \varphi) - N/C^0(s; \varphi). \quad (9)$$

Table 2 provides summary statistics.

Table 2: Summary Statistics – Baseline Profile

	Mean	Std.	Min	P25	Median	P75	Max
C (€/m)	5,528	2,297	1,560	3,598	5,608	7,459	9,369
W (€/m)	4,014	1,463	1,494	2,754	4,014	5,274	6,535
N (€/m)	3,159	1,164	1,160	2,158	3,150	4,161	5,172
RGDU (€/m)	233	213	0	108	157	357	722
A	0.591	0.058	0.550	0.558	0.562	0.600	0.748
T	0.525	0.081	0.334	0.477	0.535	0.561	0.768
ε_{NC}	0.894	0.136	0.548	0.815	0.954	1.016	1.032

Notes: Baseline analytical profile only: full-time non-cadre employee, standard territorial regime, firm with fewer than 50 employees, standard AT/MP assumptions. The full public dataset contains 521 wage points per profile and 96 profiles. $\text{SMIC} = 1,867.02$ € gross/month (June 2026). T and ε_{NC} by central differences, $h = 0.01 \times \text{SMIC}$.

4. Theoretical Framework

4.1. The Transmission Function

Definition 4.1 (Labour Cost Transmission Function). For employment profile φ , the *labour cost transmission function* is:

$$F_\varphi : C \mapsto N, \quad (10)$$

obtained by eliminating W from (1)-(2).

F_φ is well-defined (since $C(W)$ is strictly increasing), piecewise-smooth, and globally non-linear due to the RGDU structure described in Section 2.2.

4.2. Average Transmission

Definition 4.2 (Average Transmission Ratio).

$$A(C; \varphi) = \frac{F_\varphi(C)}{C} = \frac{N}{C}. \quad (11)$$

$A \in (0, 1)$ is the share of employer cost reaching the employee. In (C, N) space, $A(C)$ is the slope of the ray from the origin to $(C, F_\varphi(C))$: a declining A profile corresponds to a function that progressively departs from the 45-degree line. The complement $1 - A(C)$ is the aggregate contribution wedge; it is arithmetically equivalent to the OECD tax wedge (net of personal income tax) at any given wage point. The novelty is the functional coverage: a dense wage grid rather than 3-5 representative cases. In the current dashboard implementation, the public dataset contains 521 wage points per profile.

4.3. Marginal Transmission

Definition 4.3 (Marginal Transmission Function).

$$T(C; \varphi) = \left. \frac{dN}{dC} \right|_\varphi = F'_\varphi(C). \quad (12)$$

$T \in (0, 1)$ is the fraction of a marginal euro of employer cost that reaches the employee. By the chain rule:

$$T(s; \varphi) = \frac{1 - \tau^{S'}(W)}{1 + \tau^{E'}(W)}, \quad (13)$$

where $\tau^{E'}$ and $\tau^{S'}$ denote marginal contribution rates. In the RGDU zone, the effective marginal employer rate is elevated by the withdrawal of relief:

$$\tilde{\tau}^E(W; \varphi) = \bar{\tau}^E(W; \varphi) - \frac{d \text{RGDU}}{dW} > \bar{\tau}^E, \quad (14)$$

since $d \text{RGDU}/dW < 0$ throughout the relief zone. This directly depresses T via (13):

$$T^{\text{RGDU zone}}(s; \varphi) = \frac{1 - \bar{\tau}^S}{1 + \tilde{\tau}^E} < \frac{1 - \bar{\tau}^S}{1 + \bar{\tau}^E} = T^{\text{no relief}}(s; \varphi). \quad (15)$$

The magnitude of the depression varies within the zone. It is largest in the $[1.0, 1.6] \times \text{SMIC}$ range (steep RGCP taper, $d \text{RGCP}/dW \approx -0.71$) and smaller in the $[1.6, 2.92] \times \text{SMIC}$ range (shallower health reduction taper). Section 5 quantifies both effects.

4.4. Net-Income-to-Labour-Cost Elasticity

Definition 4.4 (Net-Income-to-Labour-Cost Elasticity).

$$\varepsilon_{NC}(C; \varphi) = \frac{\partial \ln N}{\partial \ln C} = \frac{T(C; \varphi)}{A(C; \varphi)}. \quad (16)$$

$\varepsilon_{NC} = 1$ in a proportional system ($T = A$); $\varepsilon_{NC} < 1$ whenever $T < A$ (sub-proportional, as expected in the RGDU zone); $\varepsilon_{NC} > 1$ when $T > A$ (super-proportional, when relief is being activated). The elasticity provides the cleanest summary: a 1% increase in employer cost translates into a ε_{NC} % increase in employee net income. When $\varepsilon_{NC} < 1$, the translation is less than proportional.

4.5. Relief Transmission Premium

Definition 4.5 (Relief Transmission Premium).

$$R(C; \varphi) = A(C; \varphi) - A^0(C; \varphi) = \frac{N}{C} - \frac{N}{C^0}, \quad C^0 = C + \text{RGDU}. \quad (17)$$

Since the RGDU does not affect employee contributions, $N^0 = N$, so $R > 0$ whenever $\text{RGDU} > 0$. The premium measures the percentage-point gain in average transmission attributable to the relief scheme. Its area $\int R(s) ds$ over the relief zone is a summary measure of the aggregate efficiency gain conferred by the RGDU across the wage distribution.

4.6. Framework Summary

Table 3: The FLCL Framework

Object	Expression	Interpretation
$F_\varphi(C)$	$N = F_\varphi(C)$	Full statutory mapping: employer cost $\rightarrow$ net income
$A(C)$	N/C	Average: share of employer cost reaching the employee
$T(C)$	dN/dC	Marginal: employee gain per additional euro of employer cost
$\varepsilon_{NC}(C)$	T/A	Elasticity: % net income gain per 1% increase in employer cost
$R(C)$	$A - A^0$	Relief premium: pp gain in A from RGDU

5. Results

5.1. The Transmission Function in (C, N) Space

Figure 1 plots the estimated transmission function in (C, N) space for the baseline profile.



Figure 1: Labour cost transmission function $N = F_\varphi(C)$, baseline profile. The dashed orange line is the full-transmission reference ($N = C$). The blue vertical dashed line marks the end of the RGDU analytical zone at $2.92 \times \text{SMIC}$ (employer cost $\approx 7,711$ €/month). The growing gap between F_φ and the diagonal reflects the increasing aggregate contribution wedge as wages rise; the visible change in slope near the RGDU exit reflects the step increase in employer cost as relief disappears.

The most salient feature of Figure 1 is the visible kink at the RGDU exit ($C \approx 7,711$ €/month): to the left, the function's slope is somewhat depressed by the progressive withdrawal of relief; to the right, the function reverts to a steeper, approximately linear trajectory. The vertical dashed line marks this transition clearly. The total distance below the 45-degree line at the right edge (employer cost $\approx 9,369$ €) is approximately 4,200 € – the aggregate monthly contribution wedge at $3.5 \times \text{SMIC}$.

5.2. Average Transmission: $A(s)$

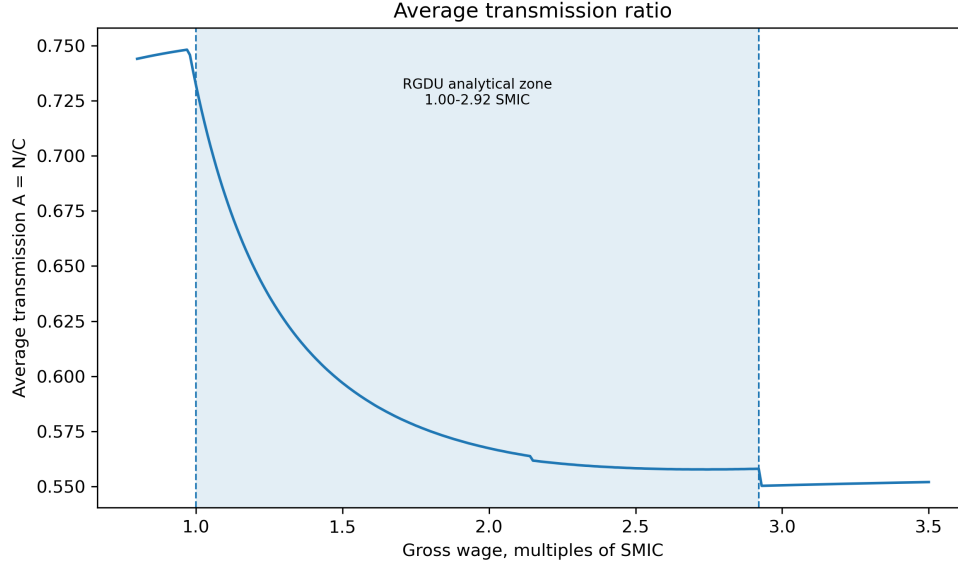


Figure 2: Average transmission ratio $A(s) = N/C$, baseline profile. Blue shading marks the RGDU analytical zone $[1.0, 2.92] \times \text{SMIC}$. The sharp initial descent from 1.0 SMIC is driven by RGCP withdrawal; the slower, continued decline from 1.6 to 2.92 SMIC reflects the residual health contribution reduction. The step discontinuity near $2.9 \times \text{SMIC}$ marks the RGDU exit, coinciding with a PSS ceiling kink.

The $A(s)$ profile exhibits two distinct phases within the RGDU zone. In the **RGCP phase** $([1.0, 1.6] \times \text{SMIC})$, A falls steeply from **0.733** to **0.588**, a decline of **14.5 percentage points** over 0.6 SMIC units – an average rate of 24 pp/SMIC. In the **tail phase** $([1.6, 2.92] \times \text{SMIC})$, A continues to fall from 0.588 to **0.558**, adding a further **3.0 percentage points** over 1.32 SMIC units – a rate of 2.3 pp/SMIC. The total A drop across the full RGDU zone is therefore **17.4 percentage points** (0.733 to 0.558). Above the RGDU exit, A is remarkably stable: it ranges only from 0.550 to 0.552 between 2.93 and 3.50 SMIC, confirming that the contribution schedule is effectively proportional in this range.

5.3. Marginal Transmission: $T(s)$

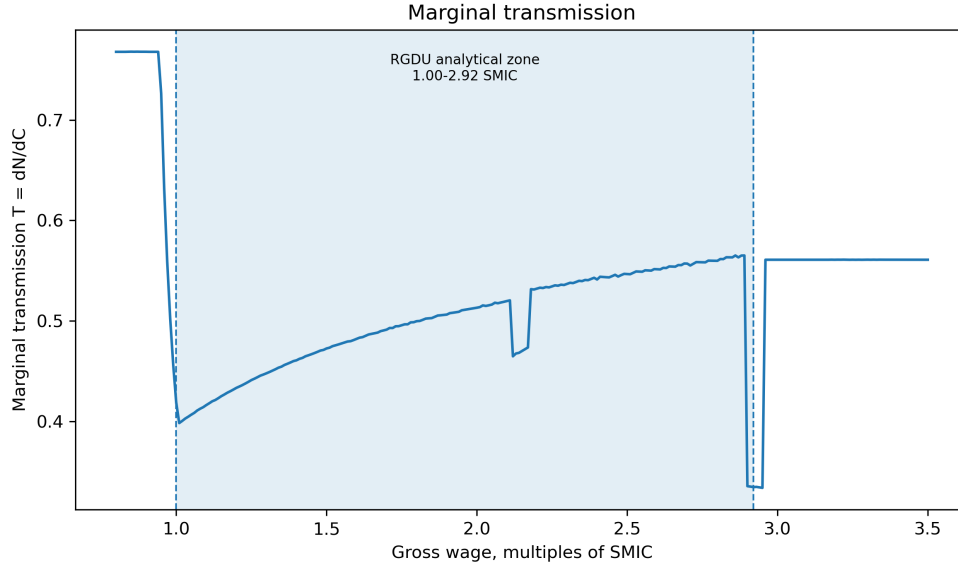


Figure 3: Marginal transmission $T(s) = dN/dC$, baseline profile. The sharp drop at $1.0 \times \text{SMIC}$ is the RGCP-driven marginal cliff; the secondary notch near $2.1 \times \text{SMIC}$ is a PSS ceiling kink; the deep trough at $2.9 \times \text{SMIC}$ marks the abrupt exit from the RGDU zone. Blue shading marks the RGDU zone.

Figure 3 is the central figure of the paper. Three features stand out.

The RGCP cliff at 1.0 SMIC. For reference, transmission below the statutory minimum wage remains high ($T \approx 0.77$), but this range lies outside the analytical RGDU zone considered in the remainder of the paper. At $s = 1.0 \times \text{SMIC}$, T drops sharply to **0.419** – the entry into the RGCP taper and the start of the analytical zone. The mechanism is captured by inequality (15): the effective marginal employer rate jumps by approximately 0.71 per unit of gross wage at the taper entry, raising the cost of delivering a marginal unit of net income.

Recovery in the RGDU zone. Within the RGDU zone, T recovers gradually as the taper slope flattens. Mean marginal transmission across the full RGDU zone $[1.0, 2.92] \times \text{SMIC}$ is $\bar{T} = \mathbf{0.498}$, compared to 0.549 above the RGDU exit – a difference of 5.1 percentage points. The recovery is faster in the RGCP phase (1.0-1.6) and slower in the tail phase (1.6-2.92): mean T in the tail phase is 0.522, versus 0.445 in the RGCP phase.

The RGDU exit trough at 2.9 SMIC. At $s \approx 2.90\text{--}2.92 \times \text{SMIC}$, T drops sharply to its global minimum of **0.334**, lower even than the RGCP cliff. This trough reflects the simultaneous coincidence of two effects: the abrupt cessation of 109 €/month of RGDU relief, and a PSS ceiling kink that alters the marginal employer contribution rate. Above

$s = 2.96$, T recovers immediately to approximately 0.561 and stabilises at that level for the remainder of the distribution.

5.4. Elasticity: $\varepsilon_{NC}(s)$

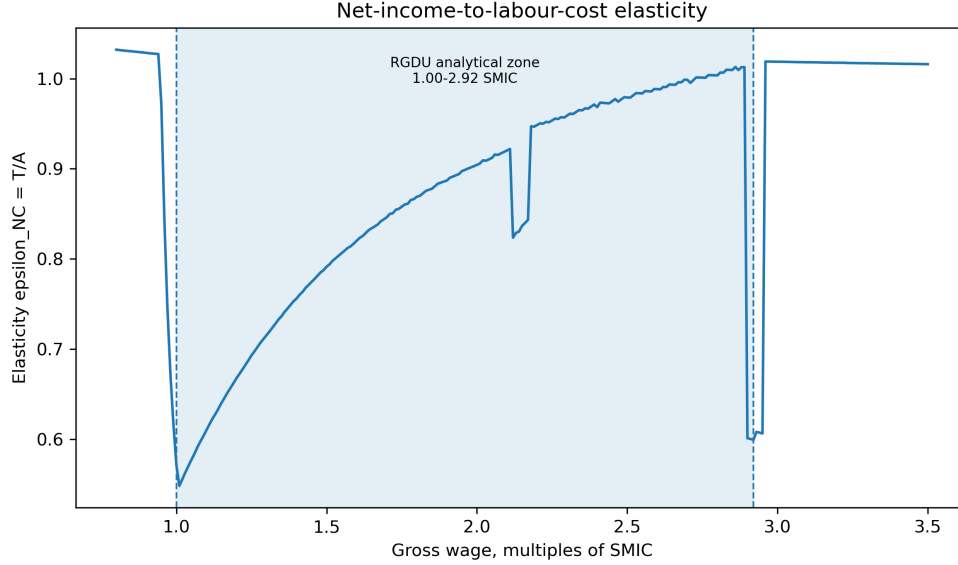


Figure 4: Net-income-to-labour-cost elasticity $\varepsilon_{NC}(s) = T(s)/A(s)$, baseline profile. The system is near-proportional ($\varepsilon_{NC} \approx 1$) below $1.0 \times \text{SMIC}$ and above $2.92 \times \text{SMIC}$, and sub-proportional throughout the RGDU zone. The minimum of ε_{NC} at $1.01 \times \text{SMIC}$ marks the point of maximum divergence between marginal and average transmission.

The ε_{NC} profile confirms the average-marginal divergence in clean, dimensionless terms. Outside the RGDU zone, ε_{NC} hovers close to 1 (range 0.99-1.03), confirming near-proportionality of the out-of-zone contribution schedule. Within the RGDU zone:

- ε_{NC} reaches its global minimum of **0.548** at $s = 1.01 \times \text{SMIC}$, immediately upon RGCP entry. At this point, a 1% increase in employer cost produces only a 0.55% increase in employee net income – a shortfall of 45% relative to the proportional benchmark.
- ε_{NC} recovers monotonically through the RGCP phase (to 0.820 at $s = 1.6$) and continues recovering through the tail phase (to approximately 1.01 near the RGDU exit).
- The recovery path is visibly concave: the elasticity gap ($1 - \varepsilon_{NC}$) closes fastest in the RGCP phase and more slowly in the tail phase, reflecting the steeper taper slope in the former.

Mean elasticity in the RGCP phase ($[1.0, 1.6]$) is **0.704**; in the tail phase ($[1.6, 2.92]$) it is **0.926**; above the RGDU exit it is **0.996**. The sub-proportional regime is therefore

concentrated in the RGCP phase, with the tail phase approaching proportionality only gradually.

5.5. Relief Transmission Premium: $R(s)$

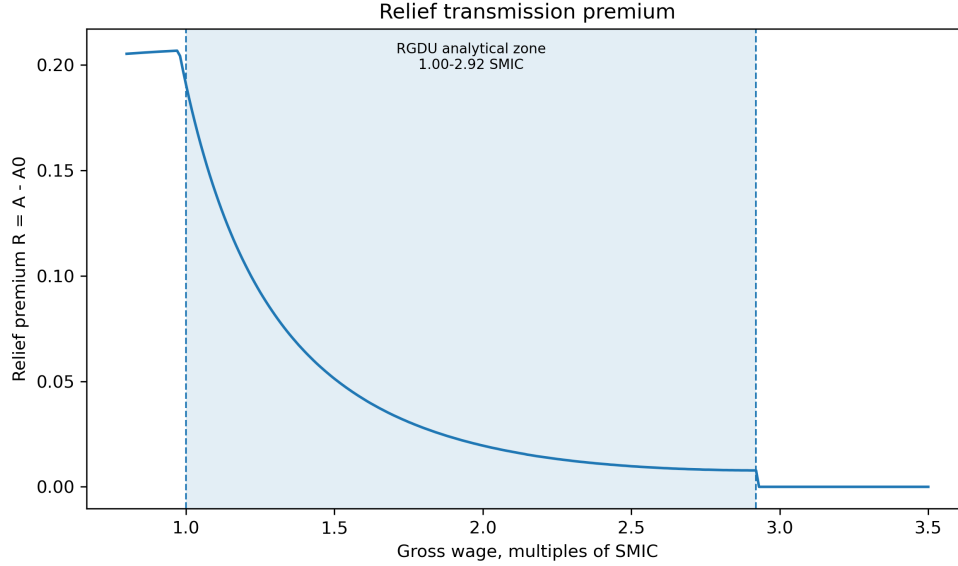


Figure 5: Relief transmission premium $R(s) = A(s) - A^0(s)$, baseline profile. The premium is non-zero throughout the RGDU zone and vanishes above $2.92 \times \text{SMIC}$. The step at $2.9 \times \text{SMIC}$ marks the abrupt RGDU exit; the small negative values immediately above the exit reflect a numerical artefact of the PSS kink coincidence.

The relief premium $R(s)$ reaches **0.191** at $s = 1.0 \times \text{SMIC}$, the entry of the analytical RGDU zone: the RGDU raises average transmission by 19.1 percentage points at the minimum wage relative to the counterfactual without relief. For reference, R is slightly higher (0.207) at $s = 0.8 \times \text{SMIC}$ (outside the analytical zone), reflecting full relief eligibility in the sub-SMIC segment of the simulation grid. Within the analytical zone, the premium *does not disappear at* $1.6 \times \text{SMIC}$: it is still 4.1 percentage points at $s = 1.6$ and remains above 1 percentage point up to $s \approx 2.2$. The area under the R curve over the analytical zone $[1.0, 2.92] \times \text{SMIC}$ equals **0.096 SMIC units**, representing the aggregate transmission gain from the RGDU over that range. Including the sub-SMIC segment $[0.8, 1.0)$ raises this to 0.118 SMIC units. Assuming instead that the relief zone ends at $1.6 \times \text{SMIC}$ would reduce the analytical-zone figure to approximately 0.050 SMIC units, an underestimate of approximately **48%** relative to the correct $[1.0, 2.92]$ boundary.

5.6. Summary of Main Results

Table 4 consolidates the key quantitative findings.

Table 4: Main Results – Baseline Profile

Statistic	Value
Minimum average transmission	0.550
Maximum average transmission	0.748
Minimum marginal transmission	0.334
Maximum marginal transmission	0.768
Minimum elasticity	0.548
Maximum elasticity	1.032
Maximum relief transmission premium	0.207
Mean marginal transmission in RGDU-positive zone	0.498
Mean marginal transmission above RGDU threshold	0.549

Notes: RGDU analytical zone defined as $[1.0, 2.92] \times \text{SMIC}$; above-RGDU threshold: $s > 2.92$.

5.7. Profile Heterogeneity and Robustness

Table 5 reports transmission statistics for the initial full-time robustness grid, covering 16 combinations of employee status, territorial regime and AT/MP scenario. The public dashboard extends this grid to 96 profiles by adding firm-size and working-time dimensions; these additional dimensions are used for interactive exploration but are not yet reported in the compact robustness table below.

Table 5: Transmission Statistics by Employment Profile

Profile	A range	T range	ε_{NC} range	$R_{\max}$	s at $T_{\min}$
Non-cadre, std, std	[0.550, 0.748]	[0.334, 0.768]	[0.548, 1.032]	0.207	2.95
Non-cadre, std, AT/MP 1%	[0.554, 0.756]	[0.335, 0.776]	[0.546, 1.032]	0.211	2.95
Non-cadre, std, AT/MP 4%	[0.543, 0.735]	[0.331, 0.754]	[0.553, 1.032]	0.200	2.95
Non-cadre, std, support	[0.556, 0.759]	[0.336, 0.779]	[0.545, 1.032]	0.212	2.95
Non-cadre, Alsace, std	[0.542, 0.737]	[0.329, 0.757]	[0.548, 1.032]	0.204	2.95
Non-cadre, Alsace, AT/MP 1%	[0.546, 0.745]	[0.330, 0.765]	[0.545, 1.033]	0.208	2.95
Non-cadre, Alsace, AT/MP 4%	[0.535, 0.724]	[0.326, 0.743]	[0.552, 1.032]	0.198	2.95
Non-cadre, Alsace, support	[0.547, 0.746]	[0.330, 0.766]	[0.545, 1.033]	0.208	2.95
Cadre, std, std	[0.545, 0.736]	[0.334, 0.755]	[0.552, 1.032]	0.201	2.95
Cadre, std, AT/MP 1%	[0.549, 0.743]	[0.335, 0.763]	[0.549, 1.032]	0.205	2.95
Cadre, std, AT/MP 4%	[0.538, 0.723]	[0.331, 0.741]	[0.556, 1.032]	0.195	2.95
Cadre, std, support	[0.551, 0.746]	[0.336, 0.766]	[0.549, 1.032]	0.206	2.95
Cadre, Alsace, std	[0.537, 0.725]	[0.329, 0.744]	[0.551, 1.032]	0.199	2.95
Cadre, Alsace, AT/MP 1%	[0.541, 0.733]	[0.330, 0.752]	[0.549, 1.032]	0.202	2.95
Cadre, Alsace, AT/MP 4%	[0.530, 0.712]	[0.326, 0.731]	[0.556, 1.032]	0.193	2.95
Cadre, Alsace, support	[0.542, 0.733]	[0.330, 0.753]	[0.549, 1.033]	0.203	2.95

Notes: “std” = standard; “support” = *fonctions support* AT/MP category. $T_{\min}$ at $s = 2.95 \times \text{SMIC}$ in all profiles, reflecting the PSS kink-RGDU exit coincidence. The RGCP-driven trough at $s \approx 1.01 \times \text{SMIC}$ is universal but is not the global minimum. $R_{\max}$ is attained at $s = 0.8 \times \text{SMIC}$ (outside the analytical zone); within the analytical zone $[1.0, 2.92] \times \text{SMIC}$, the maximum R is 0.191 (non-cadre std). The expanded public dashboard additionally covers firm-size and working-time profiles, yielding 96 profile combinations in total.

The profile comparison yields two main conclusions. First, the *level* of transmission varies predictably by profile: cadre profiles exhibit lower A and T (higher AGIRC-ARRCO tier), and Alsace-Moselle profiles are marginally below their continental counterparts (additional health contribution). Second, the *shape* of the transmission profile – the range of ε_{NC} , the location and depth of the RGCP cliff, and the RGDU exit trough – is essentially invariant across all 16 profiles. The minimum ε_{NC} ranges narrowly from 0.545 to 0.556 across profiles; the $T_{\min}$ location is identical (2.95 SMIC) in all 16 cases. The marginal cliff and the RGDU zone structure are therefore structural properties of the French contribution schedule, not profile-specific artefacts.

Table 6 provides the cross-profile zone comparison.

Table 6: Mean Marginal Transmission $\bar{T}$ by Wage Zone, All Profiles

zone	T_{mean}	<i>Notes:</i> Zones defined by the RGDU analytical zone boundary at $2.92 \times \text{SMIC}$; pooled across the initial full-time robustness grid of 16 profiles.
1.00-2.92 SMIC	0.498	
>2.92 SMIC	0.549	

6. Discussion

6.1. A Relief Zone Much Wider Than the RGCP Formula

The most institutionally significant finding of the paper is that the analytical RGDU zone extends to $2.92 \times \text{SMIC}$, not 1.6 as commonly assumed for the RGCP alone. This distinction matters for three reasons.

First, it affects the range of wages for which the implicit marginal tax on growth applies. Under the RGCP-only view, only workers between 1.0 and $1.6 \times \text{SMIC}$ face the depressed marginal transmission. The RGDU-corrected view extends this range to $2.92 \times \text{SMIC}$, covering a much larger share of the employed population.

Second, it affects the aggregate efficiency gain from the relief system. The area under $R(s)$ over the analytical RGDU zone $[1.0, 2.92] \times \text{SMIC}$ is 0.096 SMIC units. Restricting the analysis to the RGCP-only range $[1.0, 1.6] \times \text{SMIC}$ would reduce this figure to approximately 0.050 SMIC units, an underestimate of about 48% . Including the sub-SMIC segment $[0.8, 1.0]$ raises the total area to 0.118 SMIC units.

Third, it affects the location of the most severe marginal cliff. Under the RGCP-only view, the deepest cliff is at $1.0 \times \text{SMIC}$ ($T = 0.419$). The RGDU-corrected view reveals a secondary and even deeper cliff at $2.9 \times \text{SMIC}$ ($T = 0.334$), where the abrupt RGDU exit coincides with a PSS ceiling kink. This secondary trough is invisible to any analysis that limits attention to the $[1.0, 1.6] \times \text{SMIC}$ range.

6.2. Implicit Marginal Tax on Wage Growth and Policy Implications

The divergence between A and T in the RGDU analytical zone constitutes an implicit marginal tax on wage growth. For each additional euro of employer spending in the RGDU analytical zone, 50.2 cents reach the employee on average, versus 54.9 cents above the RGDU threshold – a gap of 4.7 cents per euro. In percentage terms, ε_{NC} averages 0.856 in the RGDU analytical zone versus 0.996 above it. The implicit tax is concentrated in the RGCP phase (mean $\varepsilon_{NC} = 0.704$) and much attenuated in the tail phase (0.926).

This has three direct policy implications. First, a statutory minimum wage increase that moves workers above $1.0 \times \text{SMIC}$ amplifies employer cost by the foregone RGDU relief: the effective cost per unit of net income delivered is $1/T(1.0) \approx 2.39$ euros per

euro of net income gain, versus $1/T \approx 1.82$ above the RGDU threshold. Second, firms in the RGDU zone have implicit incentives to cluster wages near $1.0 \times \text{SMIC}$ rather than raise them within the zone, since the cost-per-unit-of-net-income is higher. Third, the RGDU exit cliff at $2.9 \times \text{SMIC}$ creates a symmetric incentive near $3.0 \times \text{SMIC}$: marginal transmission recovers sharply above the exit, making wage increases just above 2.92 SMIC relatively cheap to deliver. This may create a complementary bunching incentive above the RGDU zone.

6.3. Limitations and Extensions

Personal income tax. The framework covers payroll contributions only. Incorporating PIT requires household-level information (marital status, dependants, other income). The natural extension uses the INES microsimulation model (Cabannes et al., 2014) to compute post-PIT transmission functions for canonical household types.

Statutory vs. economic incidence. The FLCL measures characterise the statutory schedule before behavioural adjustment. The question of who ultimately bears the contribution burden is the subject of the incidence literature (Gruber, 1997; Saez et al., 2019; Bozio et al., 2019); the FLCL framework provides the statutory input to those analyses.

Time dimension. All results correspond to June 2026. A time-series extension tracking F_φ , T , and ε_{NC} since the introduction of the *allègement Fillon* in 2003 would enable evaluation of the cumulative effect of two decades of labour cost policy reforms.

International comparisons. Any country with a codified payroll schedule and a public simulation engine can produce analogous transmission functions. Cross-national comparison of $T(s)$ and $\varepsilon_{NC}(s)$ profiles would provide a richer picture of labour cost incentives than OECD tax wedge comparisons.

7. Conclusion

This paper has introduced the FLCL framework, a functional approach to labour cost transmission measurement. The public dashboard implements the framework using 50,016 payroll simulations from the official Mon-entreprise / URSSAF calculation engine (June 2026), covering 521 wage points per profile and 96 employment profiles. The analytical results reported in this paper reconstruct the statutory mapping $N = F_\varphi(C)$ for the baseline full-time profile and characterise it through four objects: the average transmission ratio A , the marginal transmission function T , the net-income-to-labour-cost elasticity ε_{NC} , and the relief transmission premium R .

Three main results emerge. First, the RGDU analytical zone extends to $2.92 \times \text{SMIC}$ – nearly twice the $1.6 \times \text{SMIC}$ ceiling of the RGCP formula alone. Restricting the analysis to the RGCP-only range $[1.0, 1.6] \times \text{SMIC}$ misses the shallower but persistent tail phase and the secondary marginal cliff at the RGDU exit. Second, average transmission falls by 17.4 percentage points across the RGDU analytical zone (from 0.733 at $1.0 \times \text{SMIC}$ to 0.558 at $2.92 \times \text{SMIC}$), with 14.5 pp concentrated in the steep RGCP phase and the remaining 3.0 pp spread across the shallower tail phase. Third, marginal transmission drops sharply to 0.419 at the RGCP entry and reaches its global minimum of 0.334 at the RGDU exit, where abrupt cessation of relief coincides with a PSS ceiling kink. The elasticity ε_{NC} confirms sub-proportional transmission throughout the RGDU analytical zone (minimum 0.548), returning to near-unity only above $2.92 \times \text{SMIC}$.

These findings reveal a zone of implicit marginal taxation of wage growth extending across nearly the entire lower half of the wage distribution – a structural feature of the French contribution schedule that is entirely invisible to conventional average-based tax wedge measures.

A. Contribution Rate Schedule – June 2026

Table 7: Main Social Contribution Rates – June 2026

Levy	Employer (%)	Employee (%)	Ceiling
Health insurance	7.00	0.00	None
Family benefits	3.45	0.00	None
Old-age (uncapped)	2.02	0.40	None
Old-age (capped)	8.55	6.90	1 PSS
Unemployment insurance	4.05	0.00	4 PSS
AGIRC-ARRCO T1 (non-cadre)	4.72	3.15	1 PSS
AGIRC-ARRCO T2 (cadre)	12.95	8.64	8 PSS
CSG (deductible)	0.00	6.80	None
CSG (non-deductible)	0.00	2.90	None
CRDS	0.00	0.50	None
AT/MP (standard)	2.50	0.00	None
FNAL	0.50	0.00	None
RGCP (max, at $1.0 \times \text{SMIC}$)	−32%	0.00	Degressive, $[1.0, 1.6] \times \text{SMIC}$
Health reduction	degressive	0.00	$[1.0, \sim 2.5] \times \text{SMIC}$

Notes: PSS = Plafond de la Sécurité Sociale (3,925 €/month, June 2026). Both the RGCP and the degressive health contribution reduction are consolidated in the RGDU expression in the simulation engine. Their combined positive zone extends to $2.92 \times \text{SMIC}$ in the baseline profile.

B. Numerical Methods

All derivatives use central differences with $h = 0.01 \times \text{SMIC}$:

$$\hat{T}(s) = \frac{N(s+h) - N(s-h)}{C(s+h) - C(s-h)}, \quad (18)$$

$$\hat{\varepsilon}_{NC}(s) = \frac{\ln N(s+h) - \ln N(s-h)}{\ln C(s+h) - \ln C(s-h)}. \quad (19)$$

One-sided differences are used at boundary points. The analytical figures reported in this paper use the baseline range $s \in [0.80, 3.50]$, while the public dashboard extends the wage grid to $s = 6.00$. Integrals use the composite trapezoidal rule: $\int_{0.8}^{2.92} R(s) ds = 0.118$ SMIC units for the baseline profile. Code and dataset are available through the French Labour Cost Lab public repository.

C. Analytical Derivation: RGCP Taper Slope

Under constant $\bar{\tau}^E$, expanding equation (3):

$$\text{RGCP}(W_a) = (1 + \bar{\tau}^E) \frac{T_{\max}}{0.6} (1.6 \text{SMIC}_a - W_a).$$

The derivative $d\text{RGCP}/dW_a = -(1 + \bar{\tau}^E) T_{\max}/0.6 \approx -0.71$ is constant throughout $[1.0, 1.6] \times \text{SMIC}$. Each euro of additional gross wage in this range reduces RGCP relief by 0.71 € and raises total employer cost by $1 + \bar{\tau}^E + 0.71 \approx 2.04$ € (versus $1 + \bar{\tau}^E \approx 1.33$ € outside). The ratio $T^{\text{taper}}/T^{\text{no-taper}} \approx (1 - \bar{\tau}^S)/(1 + \bar{\tau}^E) \div (1 - \bar{\tau}^S)/(1 + \bar{\tau}^E) = (1 + \bar{\tau}^E)/(1 + \bar{\tau}^E) \approx 1.33/2.04 \approx 0.65$, broadly consistent with the empirical ratio $0.445/0.720 \approx 0.62$ in the data.

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